

3.2.5 E: Further calculus

	Content
E1	Evaluate improper integrals where either the integrand is undefined at a value in the range of integration or the range of integration extends to infinity.

	Content
E2	Derive formulae for and calculate volumes of revolution.

	Content
E3	Understand and evaluate the mean value of a function.

	Content
E4	Integrate using partial fractions (extend to quadratic factors $ax^2 + c$ in the denominator).

	Content
E6	Integrate functions of the form $(a^2 - x^2)^{-\frac{1}{2}}$ and $(a^2 + x^2)^{-1}$ and be able to choose trigonometric substitutions to integrate associated functions.

	Content
E7	Arc length and area of surface of revolution for curves expressed in Cartesian or parametric coordinates.

	Content
E8	Derivation and use of reduction formulae for integration.

	Content
E9	The limits $\lim_{x \rightarrow \infty} (x^k e^{-x})$ and $\lim_{x \rightarrow 0} (x^k \ln x)$ where $k > 0$, applied to improper integrals